

The Great Rift Valley

A system of faults in Earth's crust that extends for 3,700 miles (6,000 km) through part of southwestern Asia and then down the length of East Africa

The Great Rift Valley marks a region where plumes of hot material from Earth's mantle are welling up and causing the crust to stretch and split apart along a series of fissures, or rifts. These rifts start in southern Lebanon and pass southward, via the Red Sea, to the Afar Depression (see p.183) and finally down through East Africa, ending on the coast of Mozambique. The average width of the Great Rift Valley is around 30 miles (50km). In many parts, sections of crust have sunk down, leaving escarpments on either side, with steep

▷ SOCIAL ANIMALS

The Great Rift is home to many endemic species of African wildlife, including olive baboons. These live in groups of 15 to 150 individuals, with a complex social structure.



walls that are typically around 2,950 ft (900 m) high. The African section of the Great Rift, known as the East African Rift System, is gradually splitting the African tectonic plate in two.

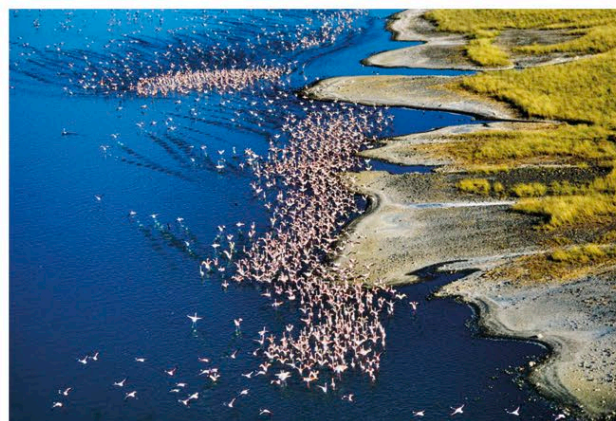
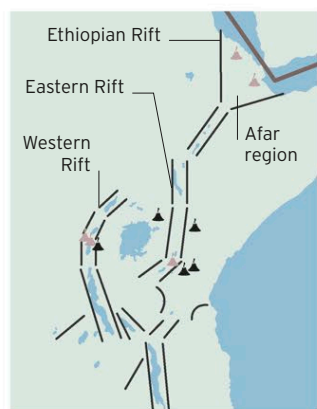
Volcanoes and lakes

Dotted throughout the rift system are numerous volcanoes, formed where magma has welled up through fissures in the crust and erupted at the surface. Several, such as Ol Doinyo Lengai in Tanzania, have been active in the past 50 years. In the East African Rift System, many lakes—some very deep—fill the valley floors. Ancient fossils of various extinct relatives of modern humans have been discovered in different parts of the region.

The Great Rift Valley has **formed over** approximately the **last 35 million years**

EASTERN AND WESTERN RIFT VALLEYS

The oldest and best defined rift occurs in the Afar region of Ethiopia. This is usually referred to as the Ethiopian Rift. In East Africa, the Great Rift splits into two branches called the Western (or Albertine) and the Eastern (or Gregory) rifts. Each contains active volcanoes and a string of lakes. Some of these are highly alkaline and are termed soda lakes; others are very deep.



△ SALINE LAKE

Kenya's Lake Bogoria is a saline and alkaline lake that is home to one of Earth's largest populations of lesser flamingos. There are also geysers and hot springs along the banks.

▷ UNIQUE LANDSCAPE

The crater of Ol Doinyo Lengai is filled with steep-sided cones and a unique dark lava that turns white as it comes into contact with moisture in the air and solidifies.

